

Practice : MST - Kruskal Algorithm

Deadline: Apr.22/23/24. Must be checked onsite (during labs).Online checks via email or messaging apps are not accepted.

1. Objective

This Practice has one question based on **Chapter 4.5 Minimum Spanning Tree** in the lecture. It is intended to help students understand the basic concepts of Kruskal Algorithm.

2. Kruskal Algorithm

General Logic

- Sort all edges by weight in ascending order
- Finding the edge(**n**, **m**) with the smallest weight
- Whether node **n** and node **m** are in a same tree?
 - If yes, skip
 - If no, merge two trees, and add the **w** value
- Stop when the MST contains **V-1 edges**

Characteristics:

Greedy algorithm, time complexity $O(E \log E)$ (dominated by sorting)

Note: Kruskal's algorithm is **not** suitable for directed graphs

Union-Find:

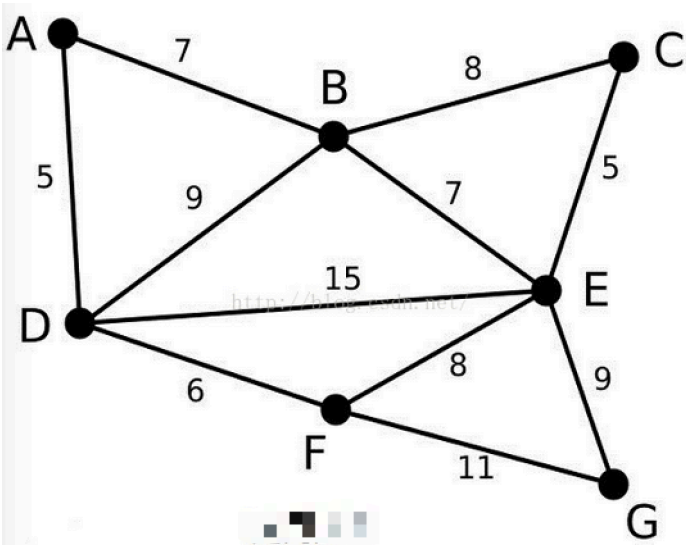
How to merge two trees (n, m)?

To check whether the two endpoints of an edge are in the same set, you can use a Union-Find data structure.

```
Find root of n and m respectively
If root of n equals to root of m, n and m is in a same tree. Skip
Else
    Get the height of root n and root m
    if(rootN.height > rootM.height) rootM.parent =rootN
    else if(rootN.height<rootM.height) rootN.parent=rootM
    else  rootM.parent=rootN    rootN.height++;
```

For example:

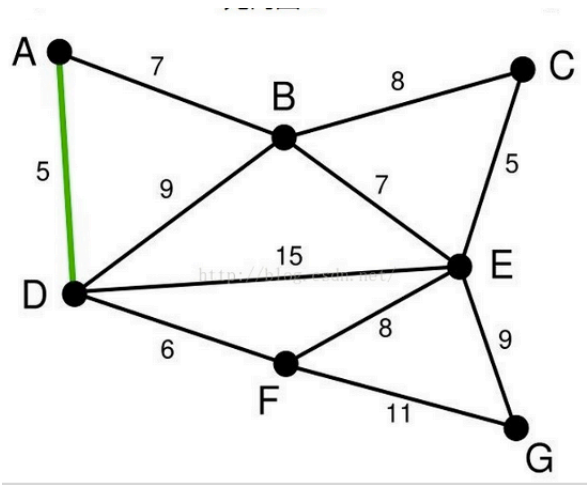
The original graph is:



The original table is:

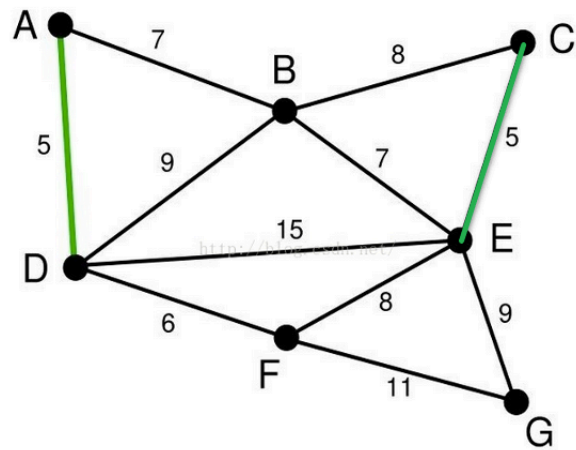
Index	0	1	2	3	4	5	6
Node	A	B	C	D	E	F	G
Parent	0	1	2	3	4	5	6
Height	0	0	0	0	0	0	0

Step 1: A-D



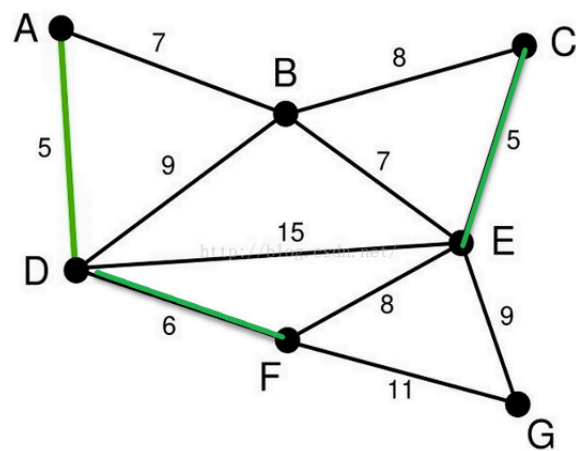
Index	0	1	2	3	4	5	6
Node	A	B	C	D	E	F	G
Parent	0	1	2	0	4	5	6
Height	1	0	0	0	0	0	0

Step 2: C-E



Index	0	1	2	3	4	5	6
Node	A	B	C	D	E	F	G
Parent	0	1	2	0	2	5	6
Height	1	0	1	0	0	0	0

Step 3: D-F

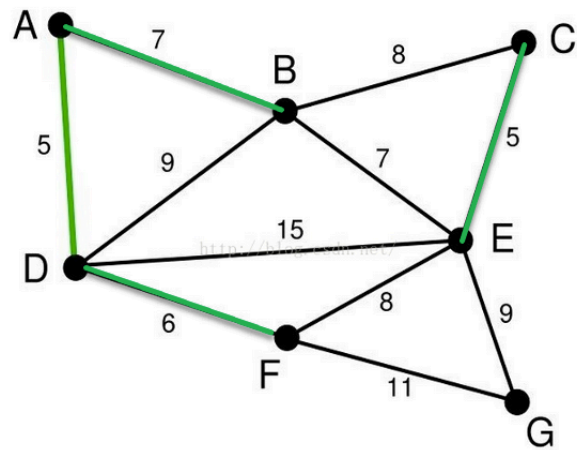


$f(\text{root}).\text{height} < d(\text{root}).\text{height}$

$f.\text{parent} = a.\text{index}$

Index	0	1	2	3	4	5	6
Node	A	B	C	D	E	F	G
Parent	0	1	2	0	2	0	6
Height	1	0	1	0	0	0	0

Step 4: A-B

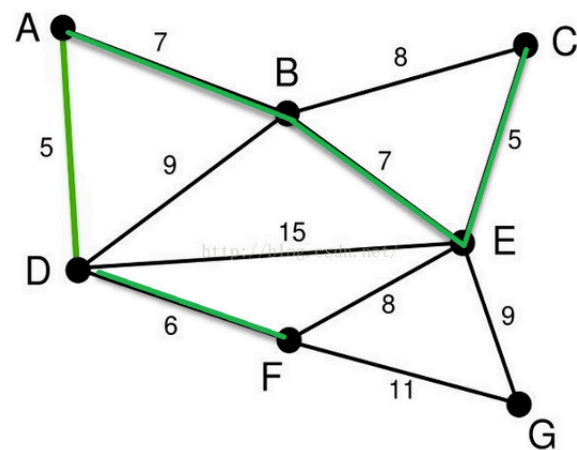


$b(\text{root}).\text{height} < a(\text{root}).\text{height}$

$b.\text{parent} = a.\text{index}$

Index	0	1	2	3	4	5	6
Node	A	B	C	D	E	F	G
Parent	0	0	2	0	2	0	6
Height	1	0	1	0	0	0	0

Step 5: B-E



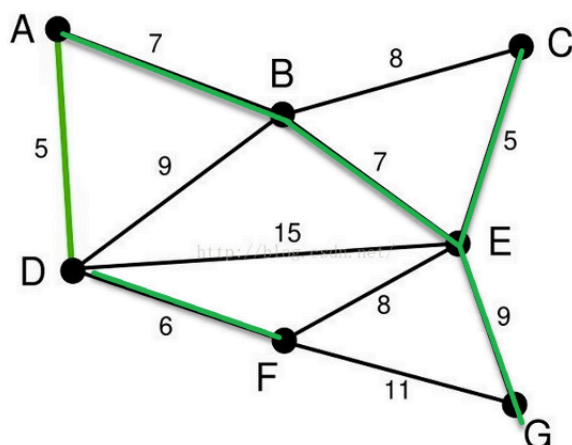
$e(\text{root}).\text{height} == b(\text{root}).\text{height}$

$c.\text{parent} = a.\text{index}$

$a.\text{height} ++$

Index	0	1	2	3	4	5	6
Node	A	B	C	D	E	F	G
Parent	0	0	0	0	2	0	6
Height	2	0	1	0	0	0	0

Step 6: E-G



$e(\text{root}).\text{height} > g(\text{root}).\text{height}$

$g.\text{parent} = a.\text{index}$

Index	0	1	2	3	4	5	6
Node	A	B	C	D	E	F	G
Parent	0	0	0	0	2	0	0
Height	2	0	1	0	0	0	0

3. Task

Description:

In **Undirected graph**, given n nodes, m edges, and the weight of each edge (weights are positive integers), Find the total weight of the minimum spanning tree.

Input Description:

The first line contains two integers **n** and **m**, representing n nodes and m edges, separated by a space.

The next **m** lines each contain three integers: the first two integers represent the node indices, and the third integer represents the **distance** between the two nodes.

Output Description:

The total weight of the minimum spanning tree

Input Test case1:

```
8 14
0 1 4
1 2 6
1 3 16
2 3 8
4 3 10
2 4 5
4 5 14
3 5 21
4 6 11
2 7 23
0 7 24
6 7 9
6 5 7
4 7 18
```

Output1

50

Input Test case2:

```
7 11
0 1 7
1 2 8
0 3 5
1 3 9
1 4 7
2 4 5
3 4 15
3 5 6
5 6 11
4 6 9
4 5 8
```

Output2

39

Input Test case3:

In the File `1.in`.

Output Test case3:

```
109545627
```

You should finish it in **1 second in Java and 0.6 second in C++**.

In this section, you can try the command:

Java

1. Mac or Linux

```
time java Main < 1.in
```

2. Windows power shell

```
Measure-Command { cmd /c "java Main < 1.in" }  
or  
(Measure-Command { cmd /c "java Main < 1.in" }).TotalMilliseconds
```

C++

1. Mac or Linux

```
time ./Main < 1.in
```

2. Windows power shell

```
Measure-Command { cmd /c ".\Main.exe < 1.in" }  
or  
(Measure-Command { cmd /c ".\Main.exe < 1.in" }).TotalMilliseconds
```

4. Submission

Show the following to your tutor or SA **onsite**:

Grading:

- Submissions on time:
 - 0.5 points for first two test cases.
 - 0.3 points for the File `1.in`.
 - 0.2 points for finish it **in time limited**, and your algorithm of **Union-Find**
- Late submissions: 0 points.

